Introduction

The classification of living organisms has been controversial throughout time. Aristotle's system distinguished only between plants and animals on the basis of movement, feeding mechanism, and growth patterns. In 1735 the Swedish naturalist Carolus Linnaeus formalized the use of two Latin names to identify each organism, a system called binomial nomenclature. He grouped closely related organisms and introduced the modern classification groups: kingdom, phylum, class, order, family, genus, and species. Single-celled organisms were observed but not classified. In 1866 German biologist Ernst Haeckel proposed a third kingdom, Protista, to include all single-celled organisms. Some taxonomists also placed simple multicellular organisms, such as seaweeds, in Kingdom Protista. Bacteria, which lack nuclei, were placed in a separate group within Protista called Monera. In 1938 American biologist Herbert Copeland proposed a fourth kingdom, Monera, to include only bacteria. This was the first classification proposal to separate organisms without nuclei, called prokaryotes, from organisms with nuclei, called eukaryotes, at the kingdom level. In 1957 American biologist Robert H. Whittaker proposed a fifth kingdom, Fungi, based on fungi's unique structure and method of obtaining food. Fungi do not ingest food as animals do, nor do they make their own food, as plants do; rather, they secrete digestive enzymes around their food and then absorb it into their cells. Below is the present classification system currently in use:

Monera	Prokaryote) Unicellular
Protista	Eukaryotes) unicellular
Fungi	Multicellular
Plantae Green	Multicellular
Animalia	Multicellular Animals

Plant Kingdom

The kingdom Plantae accounts for the largest proportion of the earth's biomass with its approximately 250,000 species of mosses, liverworts, ferns, flowers, bushes, vines, trees, and other plants. Aquatic and terrestrial plants are the basis of all food webs. They Contribute life-supporting oxygen to the atmosphere and provide humans with the fossil fuels, medicines, and other substances so important to our present existence.

Plant Classification

The first scientific study of plants was the attempt to classify them. At first, because of the limited knowledge of plant structures, artificial classifications, beginning with the most ancient one into herbs, shrubs, and trees, were necessary. These simple categories merely cataloged, in a tentative way, the rapidly accumulating material, in preparation for a classification based on natural relationships. Modern taxonomic classification, based on the natural concepts and system of the Swedish botanist Carolus Linnaeus, has progressed steadily since the 18th century, modified by advances in knowledge of morphology, evolution, and genetics. Thus, taxonomy is concerned with the description, identification and naming of plants and their classification into groups based on resemblances and differences mainly in their morphological, reproductive and anatomical characteristics. The objectives is to describe, name and classify plants is such a way that their evolutionary trend is brought out i.e. from simpler, earlier and more primitive types to more complex, more recent and more advanced types. The earlier classification of plants were based on either economic value e.g. cereal, medicinal, fibre yielding, oil yielding fruits, vegetables etc or on gross structural resemblance e.g. herbs, shrubs trees, climbers etc. Plants that do not fit into such classification were usually ignored

Types of system of classification

There are two Types of system of classification, these are: Artificial system of classification and Natural system of classification.

Artificial system of classification

In Artificial system of classification, one or few characters are arbitrary selected and plants are arranged into groups according to these characters. Related plants are often placed in different groups. Unrelated plants are put in same group because of the absence or presence of a particular character. This system does not indicate the natural relationship that exist among the individuals of a group. Nevertheless, the identification of an unknown plant is rendered much easier by this system.

Natural system of classification

Michiel Andarson (1727 – 1806) is the first person to reject all the artificial systems and propose his classification in favour of natural system. They devided seed plant into subkingdom **cryptogamia** (Algae) and **phanerogamia** (Gymnospermae and Angiospermae), Angiospermae (Dicotyledonae, monocotyledon), Dicotyledon (polypetalae, gamopetalae, monochlamydae), polypetalae (thalamiflorate, disciflorae, catyciflorae), gamopetalae (inferae, heteromerae, bicarpelliatae). In Natural system of classification all important characters are taken into consideration. Initially, only morphological and anatomical characters are considered and plants are grouped according to these characters. Later, cytological, reproductive, physiological, genetic characters are taken into consideration including the geographical distribution of the plant. Plants are first classified into few large groups. These are further divided and sub-divided into smaller groups until the smallest group is reached (i.e species). This system gives a true idea of natural relationship existing between the different plants. The natural system has two main aspects i.e Systematics and Nomenclature.

Systematics

This is concerned with putting plants into groups. The smallest group is the ***"SPECIES'*** which is defined as a group of individuals having a close resemblance with one another structurally and functionally. The members belonging to the species can interbreed in nature freely and successfully among themselves to give rise to progeny of the same kind. However, there is no sharp demarcation between allied species. The collection of species which bear close resemblance to one another form a ***GENUS.*** For example, Fig (*Ficus carica*), Banya (*Ficus bengalensis*) and Peepul (*Ficus religiosa*) are different species but are allied because they resemble one another in their reproductive organs-inflorescence, flower, fruit and seed. The genus name is ***Ficus.*** The groups of genera which show general structural resemblance to one another form a ***FAMILY***. For example, the genera *Gossypium, Hibiscus, Thespesia, Sida, Abutilon* and *Malva* belong to the family *alvaceae*. The *families* that are similar are grouped into an ***ORDER***. *Orders* that are

similar are grouped into a ***CLASS.*** *Classes* that are similar are grouped into a ***DIVISION.*** And *Divisions* that are similar are grouped into ***KINGDOM*** i.e ***PLANT KINGDOM***.

According to this system, plant kingdom is divided into three divisions namely *BRYOPHYTA PTERIDOPHYTA* and *SPERMATOPHYTA*. However, there is no uniformity in size of categories. Sometimes there are sub-categories such as sub-division, sub-class, sub-order and sub-family.

Nomenclature

This is concerned with naming of groups and the individuals of a group. Generally, the plant name has two parts. The first part is the *Generic* name i.e *Genus* to which the plant belongs. The second part is the *Specific* name i.e *Species* to which the plant belongs. This system is known as Binomial System of Nomenclature. It was first established by Carolus Linnaeus and later settled by the International Botanical Congress in 1932. Attempts have been made to establish a universal name endings for the higher categories. *Class* names usually ends in–aceae for example *MYXOPHYCEAE*. *Family* names end in-ae for example, *Compositae*, *Palmae*; *Orders* end in-ales for example *Filicales*. *Divisions* end in – phyta for example *Spermatophyta*. There are rules guiding the naming of plants, and this includes:

1. Generic name starts with a capital letter and specific name starts with a small letter, example *Mangifera indica*.
2. No two plants should have the same botanical name
3. Where plant has more than one name, the first name is the valid.
4. Names are underlined separately if written or typed
5. Names are printed in italics
6. Name of author is abbreviated, and it is found after the name of *Species* example *Mangifera indica, Linn.*

KINGDOM PROTISTA

This consists of unicellular, eukaryotic organisms such as algae, like volvox, spirogyra etc. The group algae have the following characteristics:

- Thalloid body not differentiated into root, stem and leave with little cellular differentiation.
- Contain green pigment (chlorophyll)
- Some algae contain other pigments like blue, yellow, red, brown etc in addition to the chlorophyll.
- The exibit Autotrophic mode of nutrition
- Store excess glucose in form of starch
- Reproduction is by fragmentation spore formation or conjugation
- Cell wall made up of cellulose.
- Body made up of true parenchymatous cells
- Some are unicellular, others are multicellular the multicellular species could be filamentous or colonial
- They are mainly aquatic and are considered as the most primitive of plants
- Example of plants in this group include: Blue green algae (*Nostoc, Anabaena,*)
- Green algae(*Volvox,Spirogyra,Cladophora,Chara,Nitella*)
- Yellow green algae (*Vaucheria*) Diatoms,

Classification of Algae (Phycophyta)

Below are the structural Characteristics and examples of members of the Division Phycophyta

- Mostly are celled algae
- Posses variety of forms
- Single cells, may form filaments or colonies

- Found in fresh as well as salt water and in wet grounds
- Boat shaped, rod-shaped, disc-shaped wedge-shaped, oval-shaped, rectangular shaped e.t.c

In some *Genera*, there are ingrowths of walls which are called central or polar nodules according to position. A line called raphe extends from central to the polar, nodules. The protoplasm consists of a thin layer of cytoplasm within the cell wall. One large or many small yellow to golden-brown plastids of varied shapes and sizes, a central nucleus suspended by cytoplasmic threads or by a broad cytoplasmic band, and a central vacuole. Gold brawn colour of plastids is due to a pigment called diatomin in addition to chlorophyll. Pyrenoids may or may not be present. If present, are without the starch envelop. Reserve foods are globules of fats and insoluble complex substance called volutin. No starch in diatoms

The division Phycophyta is divided into the following classes

1. **Class: Myxophyceae or Cynophyceae (Blue-green algae)**

Are small group of primitive algae characterized by the presence of a blue pigment, phycocianin, in addition to chlorophyll they mainly grown in fresh water and wet ground surfaces are often found in plenty in almost every stagnant pool of water. From the cytoplasmic background lack of organize protoplast (cytoplasm, nucleus and plastids) i.e. prokaryotic, no flagella, storage compounds are Cyanophycin, oil and glycogen. No sexual reproduction but possibly genetic interchange between individuals. Asexual reproduction by a mitotic cell: fragmentation, non-motile resting spores, vegetative cells.

Example, Nostoc, Oscillatoria, Anabaena.

2. Class: **Euglenophyceae:** These are simple organism from which evolution of other plants has started example Euglena.

3. **Class: chlorophyceae (green).**

This alga are characterized by the presence of chlorophyll located in definite plastids (Chloroplasts) they are Mostly fresh water algae, but some species are terrestrial and not few are marine. Green algae exhibit a variety of forms: (a) unicellular or colonial, being motile or non-motile (b) Multicellular, being thaloid or filamentous and (c) Coenocytic.e.g. *Volvox, Spirogyra Chlamydomonas, Pandorina.*

4. **Bacillariophyceae (diatoms).**

marine fresh water and any damp surface. Constitute much of the phytoplankton, free floating and epiphytic, unicellular or in chain or in group of cells unique wall form in 2 overlapping halves (Petri dish) silicified, sculptured, eukaryotic food stored as oils and leucosin. Asexual reproduction by cell division in which generation became successively smaller Mostly free-Floating. Boat, rod, disc-shaped.

5. **Phaeophyceae (Brown).**

Entirely marine, eukaryotic form range from fixed branched filamentous to partially organized expanded thallus food reserve are laminarin and mannitol Show alternation of generation Possess brown pigment (fucoxanthin) example *Focus, Laminaria Ectocarpus and Sargassum*

6. **Rhodophyceae (red).** Possess red pigment-phycoerythrin example *Polysiphonia, Batrachospermum..*

FUNGI (Mycophyta)

This division has the following characteristics:
- They are mainly terrestrial but the lower forms are found wide spread in aquatic environment.
- Body is composed of thread like structure called hyphae (collectively the filamentous fungal body is called MYCELIUM)

- Body not differentiated into roots, stem and leaves
- Lack chlorophyll, the colour may be due to pigmentation in cell walls, cytoplasm or oil globules in the cells.
- Exhibit heterotrophic mode of nutrition
- They can be either parasitic or saprophytic but some form symbiotic association with higher plants (*Mycorrhizae*) and some form natural unions with algae to form (*Lichens*)
- Excess sugar stored in form of glycogen
- Body is made up of pseudoparenchymatous tissues
- Cell call made of cellulose mixed with chitin.
- Contain other pigment in cell wall or cell cavity.
- Almost all show an asexual or imperfect spore stage and many reproduce sexually at a "perfect stage" through true sexual fusion of gametes.
- They have a variable life cycle
- Example slime fungi, Pythium, Albugo, Mucor Rhizopus Yeast Agaricus, Puccinia.

Classification of Fungi

Division Myxomycota

Classes

i. Acrasiomycetes
ii. Hydromyxomycetes
iii. Myxomycetes
iv. Plasmodiophoromycetes

Division Basidiomycota

classes

i. Teliomycetes
ii. Hymenomycetes
iii. Gasteromycetes

Division deuteromycota

classes

i. Blostomycetes
ii. Hyphomycetes
iii. coelomycetes

The division Mycophyta is divided into 5 classes.

1. Class: Myxomycetes (slime fungi).
- Grow in damp, shady places in soil such in humus
- Require moisture and darkness for normal growth
- Contain pigments like yellow, orange, brown and red or colourless
- Animal like in their vegetative stages and plant like in their reproductive stages.

2. Class: Phycomycetes
- Many are aquatic or damp loving forms, fewer fully terrestrial
- Mycelium is unseptate and coenocytic
- Sporangium has many sporangiospores
- Sexual reproduction is either oogamous or isogamous.
- Asexual reproduction by zoospores, conidiospores or sporangiospore
- Cell wall often cellulose food reserve is oil glycogen

- Example pythium, phytophtora, mucor, albugo etc
Order I saprolegniales example saprolegnia
Order II peronosporales example pytopthora, albugo
Order iii mucorales example muco, rhizopus

3. Class: Ascomycetes
- Terrestrial, parasitic or saprophytic
- Often highly specialized
- Mycelium is of septate hyphae
- Protoplast is uninucleated or binucleated
- Asexual reproduction by non motile spores usually conidiophore
- Reproductive structure are asci

e.g Sacchromyces, Penicillum, etc
order I Sacchromycetes example yeast
order ii Aspergillales example penicillium and asperligilus
order iii Erysphales example erisphe and uncinula
order iv Sphaeriles example neurospora and xylaria
order v Hypocreales example claviceps
order vi Pezizales example peziza and ascobolus

4. Basidiomycetes (club fungi)
- Saprophytes or parasites
- Often highly specialised
- Mycelium is of septate hyphae
- Cells are usually binucleate with haploid nuclei
- Asexual reproduction by conidiospores

Example Agaricus, puccinia, ustilago
Order I Ustilaginales example ustilago
Order ii Uredinales example puccinia
Order iii Agaricales example mushroom (agaricus, amanita, etc) coral fungi (clavaria) and various form of pore fungi (polyporus, polystichus, forms, boletus, etc
Order iv Lycoperdales example puff ball (lycoperdon)
Order v Phallales example stinkhorns (phallus)
Order vi Nidulariales example birds (nidularia and cyathus)

5. Deutromycetes (fungi imperfecti)
- Saprophytes or parasites
- Reproductive structures are the conidia
- Have imperfect life-history

e.g Fusarium, Helminthosprium

MAJOR GROUPS OF PLANT KINGDOM

There are different varieties of plants on earth and these plants have been classified in to three major groups.Viz:
1. Bryophyta (liverworts & mosses)
2. Pteridophyta (ferns)
3. Spermatophyta (seed producing plants)

Division: Bryophyta

This group has the following general characteristics:

- Body divided into root, stem and leaf like structures. The root like structures are called RHIZOIDS
- Dichotomous branching is common
- Contain photosynthetic pigment similar to that of higher plants chlorophyll.
- Well defined sexual reproduction with clear cut heteromorphic alternation of generation.
- Mainly terrestrial, few are aquatic
- Low degree of tissue differentiation
- Persisting plant body is the haploid gametophyte, diploid sporophyte is always wholly or partially dependent on the gametophyte.
- Spores are always of one kind i.e homosporous.
- The spores germinate in to a structure called protonema

Classification of Bryophyta

The division bryophyta constitute the non vascular plants and is divided into 3 classes

1. Class Hepaticae (liverworts)

Gametophyte is dorsovetral, prostrate and thalloid in nature

- Dichotomous branching is common
- Growth is from apical cell or row of cells
- Sporophyte is sterile tissue, no chlorophyll and in dependent on the gametophyte example marchantia, porella, pellia

2. Class: Anthocerotae (horned liver wants)

- Gametophyte is simple and thalloid
- Sporophyte is complex
- E.g anthoceros

3. Class: Musci (mosses)

- Gametophyte is erect (in most cases) differentiated into an axis with leaf like structures above ground.
- Growth is from a single apical cell
- Sporophyte show a high degree of differentiation
- Contains chlorophyll and possess of stomatal aperture

e.g funaria, sphagnum, polytrichum.

Morphology of Marchantia

- Plant is thalloid with dichotomous branching
- Dorsoventral symmetry. Has distinct mid-rib
- Grows on damp ground or old walls forming a green carpet
- The thallus beans on it ventral side a unicellular rhizoids, a raw of scales along the midrib, 2 or 3 raws of scales on each side of the mid-rib with the outer row being close to the margin of the thallus
- On the dorsal side, there are a number of cup-like outgrowths called GEMMA cups on the mid-rid.
- Male plant beans special reproductive branches called antheridiophores with a disc or receptacle on top
- Female plant beans the archegoniophores with disc or receptacle on top.

Division Pteridophyta

This group has the following general characteristics:

- Body divided into root, stem and leaf system
- Have high degree of internal tissue differentiation with conducting system of xylem and phloem
- Dichotomous branching is common
- Growth from apical cell
- Contain photosynthetic pigments-chlorophyll a and b with caretonoids
- Clear – cut heteromorphic alternation of generation
- Spores may be homosporous (Identical) or heterosporous (of two kinds)
- Well defined sexual reproduction with antheridia (male gametes) and Archegonia (Female gametes)
- Persisting plant body is the Diploid sporophyte which is independent of the haploid gametophytes

Classification of pteridophytes

The division Pteridophyta constitute the vascular plants and is divided in to the following classes:

Class: Lycopsida

Members of this class are of worldwide distribution with some fossil forms and a few living forms. The sporophyte is differentiated in to root stem and leaf but the leaves are always small in relation to stem size (Microphyllous).

There is no leaf gap in the stele. There are homosporous and heterosporous representatives. The sporangia aggregated in to cones or strobila usually at the ends of branches example Lycopodium, Selaginella.

-- Mainly terrestrial with few aquatic members

1. Class Pteropsida

These are also of word wide distribution with many living and fossil forms. The sporophyte is differentiated in to roots, stem and leaves may be lacking in mature plants. Leaves always large in relation to stem and spirally arranged. Leaf gaps in the stele in most cases. They are Homosporous except in one family (Salviniaceae). Sporangia borne on margins or abaxial faces of leaves, usually in clusters or sori, except in one group of fossil forms, example Dryopteris.

2. Class Psilotopsida

The sporophyte is rootless and the shoot is composed of branches only or with small leaves spirally arranged. There is no leaf gaps in the stele. They are Homosporous and the sporangia is borne singly at the tips of branches. Example Whisk ferns.

3. Class: Psilophytopsida

These class comprises of only fossil order

4. Class: Sphenopsida

The sporophyte is differentiated in to roots, stem and leaf. The leaves are nearly always microphylous and arranged in whorls. There is no leaf gaps in the stem. The sporangia is Homosporous and borne on distinctive sporangiospores in a strobilus, example Equisetum.

Structure of selaginella

The saprophyte body is composed of a creeping stem bearing branches more or less regularly spaced. The stem braches develop at much the same rate as the main axis, therefore dichotomy appears to be the case. But in reality, the braches are lateral in development and the branching system is maopodial. The creeping habit keeps all the stems more or less in a horizontal plane and the leaves also expand in this plane to form a markedly dorsal ventral arrangement. The leaves are produced in four rows and are placed in opposite pairs, each pain having one large and are small leaf. The larger leaves originate from the lower lateral surface of a stem and the smaller ones from the upper surface. Each leaf bears a small membranous outgrowth called a ligule which is recognizable in young leaves but which withers and disappears on ageing. Slightly behind each fork in the stem, sooner or later a structure resembling a root develops and grows species, a large tuft of roots may developed at the tip of the rhizospore. A mature fertile plant bears vertical branches which differ from the rest of the stem structure. They are the strobili, bearing sporophylls all of equal size in four vertical rows. In the axils of the sporophylls, the sporangia are developed.

DIVISION SPERMATOPHYTA

This group has the following general characteristics:
- Body differentiated into root, stem and leaves
- Dichotomous branching is rear
- Show highest degree of internal differentiation
- Growth is from apical meristem
- Secondary thickening is common
- Contain photosynthetic pigment-chlorophyll a and b with caretonoids
- Clearly defined alternation of generation
- Well defined sexual reproduction and fertilization by means of a pollen tube
- Produce seeds which may be borne naked or in a closed structure (fruit)
- Mainly terrestrial but with some obviously derived aquatic representatives, example Cycas, Maize, Rose etc.

Classification of spermatophyta

Divided into 3 classes

1. Pteridospermae (seed-bearing ferns)
- Have vegetative characteristics of pteridophytes but bear seed-like reproductive structure
- Farms a link between true ferns and the flavoring plants
- All are extinct (No longer in existence)

2. Gymnospermae
- Mostly trees and shrubs
- Xylem element always tracheids except one order primitive vessels are found.
- Flowers are mostly of strobiloid construction and unisexual
- Seeds with one integuments and naked e.g Cycas, pins, Taxus

3. Angiospermae
- World's dominant vegetation in all habitats
- Adopted to life on land
- Posses true xylem vessels
- Flower consisting of works of sterile and fertile parts.
- Flowers usually hermaphrodite
- Male gametophyte represented by pollen grain, female by embryo sac with of nuclei
- Endosperm formed as a result of triple fusion
- Seed has two integuments enclosed in specialized capels called ovary which ripens into a true fruits.

Classes of Gymnosperms.

Gymnospermae are divided in to three classes, these are:

1. Class: Cycadopsida, example Cycas
2. Class: Coniferopsida example Pinus, Ginkgo
3. Class: Gnetopsida example Gnetum, Ephedra

Classes of Angiospermae

Sub-class monocotyledon

- Mostly herbaceous, few palm but no tree
- Adventitious root system
- Usually parallel veined leaves
- Vascular bundles irregularly arranged in the stem.
- Secondary growth is rare
- Flaver parts are in threes or multiple of threes
- Single cotyledon in the seal
- E.g maize wheat

Sub – Class Dicotyledon

- Mostly herbs, shrubs and trees
- Tap root system
- Vert-veined leaves
- Vascular bundles arranged in rings in the stem
- Secondary growth is usual
- Floral parts are in fours or fives or multiple of these.
- Two cotyledons in the seed (reely one by reduction).

The structure of a gymnosperm pinus plant

It is a tall, erect, evergreen trees.

- can grow up to a height of 45 metres.
- Has well developed tap-root and numerous aerial branches with needlelike leaves
- Stem is rugged and covered with scale bark
- The branches are of two kinds:

1 Long branches (of unlimited growth) in apparent wherls developing from lateral buds and
2 Dry branches (of limited growth).

- Leaves are also of two kinds:-

1 Long, green, needle-like foliage leaves, called needle borne only on branches (foliar spurs) and
2. Small, brawn, scaly leaves borne on both kinds of branches. The number of needles (leaves) in a cluster varies from 1 to 5 depending on the species

Structure of Angiosperms

- Range in size from minute forms e.g Wolffia to giant trees
- Flavor, plants as the higher seed plant
- Ovule matures into a seed and is produced within the space enclosed by a carpet thus are sand to produce enclosed seeds
- Matured ovary or fruit exhibit a wide variety of forms. May be simple or compound fruits, dry or flashy fruit
- Leaves are megaphylls, large and broad

haploid (n) Endosperm is formed from definitive nucleus only after fertilization and is triploid (3n) cotyledons are 1 or 2.

DIFFERENCES BETWEEN GYMNOSPERM AND ANGIOSPERM

GYMNOSPERM

ANGIOSPERM

Common flowering plant families some monocotyledonous plant families

Characteristics of the family gramineae (poaceae)

Habit – Herbs, rarely woody (bamboos)

Stem – Cylindrical with distinct nodes and internodes sometimes hallow

Leaves – Simple, alternate, with sheathing leaf-base split on the side opposite leaf-blade

Flowers- Usually bisexual, sometimes unisexual monoecious.

Perianth – Represent by two or three minute scales (lodicules)

Androecium – Stamens three, sometimes to, anthers versatile and pendulous

	GYMNOSPERM	ANGIOSPERM
1.	Xylem exclusively made tradechied Xylem mainly composed of resin	Xylem exclusively made tradechied Xylem mainly composed of resin
2	Phloem contain no companion cells	Phloem contain companion cells
3	Flavers are simple no calyx no corolla	Flavers are complex
4	Flowers always unisexual	Flavers unisexual bisexual
5	Air-current is the only pollinating Agent	There are many different pollinating agents
6	Ovules are freely exposed on the megasporohyll (carpel)	Ovules remain enclosed in the ovary.
7	During pollination, pollen grain enter the micropyle and are lodged on the nucleus	Pollen grain are deposited on the stigma
8	Male gametophyte presented by a fewcells usually 2 or 3 a vestigial	Male gametophyte reduced to 2nuclei only i.e tube nucleus and generator nucleus
9	Female gametophyte is a large structure with archegonia embedded in it, each with on ovum	Female gametophyte is represented by an 8-nucleate embryo – sac and the ovum is free in it without any archegonia.
10	Endosperm formed from vegetative tissue of female prothallus before fertilization and is haploid (n)	Endosperm is formed from definitive nucleus only after fertilization and is triploid (3n) cotyledons are 1 or 2.

Gynoecium – Carpe (3), reduced to one (according to same authors) by fusion or suppression of two, ovary superior, one – celled, with ovule, styles two (but three in baboos, and two fused into one in maize), rarely terminal or lateral stigmas feallery

Fruit – caryopsis. Pollination by wind. e.g Grass, Bamboo.

Family – Palmae

Habit – Shrubs or tree, sometimes climbing

Stem- Erect, unbranched and woody. Has scars or shed leaves. Height 45m above

Roots – Numerous, fibre developing from base of stem.

Leaves – Usually forming a crown, petiole with stout sheathing base.

Flowers – sessile, small and inconspicuous but produced in large numbers (12,000 indicate palm regular, hypogenous, unisexual (rarely bisexual), male and female flowers in some inflorescence or in the either monoecious or dioecious.

Perianth – in two whorls, 3 + 3 the outer being smaller, usually free, persistent in the female flower

Androecium – stamens usually in two whorls 3 + 3 filaments free or connate anthers versatile and two celled

Gynoecium carpels (3) OR three ovary superior, unilocular aartiloculer with one or three ovules

Fruit – drupe, berry or nut. Pollination by wind, sometimes insects Floral formulae

Family Liliaceae

Habit – herbs, climbers, rarely shrubs and trees with a bulb or rhizome or with fibrous roots

Leaves – are simple, radical or cauline or both (radical = cluster of leaves arising from short underground stem an it from the roots cauline = directly borne by the aerial parts of the stem and branches)

Flowers – regular, bisexual (rarely unisexual, dionegroud three – mercus and hypogenous, bracts usually small, scarious (thin, dry and membranous)

Perianth – Petals are petaloid, usually 6 in 2 whorls 3 + 3 free or (3 + 3) united (i.e polyphyllous and gamohylous respectively)

Androecium – stamens are 6 in 2 whorls, 3 + 3 rarely 3, free or united with perianth at the base (epiphyllous), anthers often dorsifixed

Gynoecium – carpels (3) syncarpous, ovary superior, ovules usually (infinite)
in two rous, placentation axile, styles (3) or 3.

Fruit – a berry or capsule.

Dicotyledonous Plants

Characteristics of some dicotyledonous Families

7.2.2 Family Leguminosae

Habit: Herbs, shrubs, trees, twiners, climbers

Leaves: Alternate, Pinnately Compound, rarely simple, stipules 2, usually free

Flower: Bisexual and complete, regular or irregular (actinomorphic or
Zygomorphic) hypogenous or slightly perigynous

Calyx: Sepals usually five with the cold one anterior (i.e away from the axis sometimes

Carolla: Petals usually five, with the odd one posterin

Androecium: Stamens usually ten or numerous, sometimes less than 10 by reduction, free or united

Gynoecium: Carpel one, ovary one-celled with one to many ovules, superior, placentation marginal

Fruit: commonly a legume or pod (dehiscent), sometimes lomentum (in dehiscent)

The family is large and is divided into three – subfamilies Subfamilies Caesalpinioideae
Mimosoideae Faboideae

Scientific classification

Kingdom: Plantae

Division Magnoliophyta:

Class: Magnoliopsida

Order: Fabales

Family: Fabaceae

Family: Malvaceae

Habit: Herbs shrubs, trees
Leaves: simple, alternate and palmately viewed
Flowers: Irregular, Polypetalous, bisexual, hypogenous, copiously mucilaginous with a whorl of epicalyx
Calyx: Sepals (5), valvete
Carolla- petals 5, attached to the base of staminal tube
Androecium: Stamens numerous, monadelphous (united into one bundle = staminal column/tube), epipetalous (staminal tube adnate to the petals at the base) anthers reinform, unilocular, pollen grains large and spiny.
Gynoecium: Carpels commonly (5 to α), ovary superior, 5 loculus placentation axile, style passes through staminal tube, stigma is free as many as the carpels
Fruit: Capsule or schizocarp
Seed: Endospermic

Family: Compositae e.g Tridax

Habit: Herbs and shrubs, rarely twiner or tree
Leaves: Simple, alternate or opposite, rarely compound.
Flowers: (or flared) are of two kinds. Central ones called disc florets. They are tubular and the marginal ones called ray florets. They are ligulate sometimes all florets ar of the same kind (tubuler or ligulate)
Disk Floret: Regular, bisexual, epigenous
Calyx:- Modified into hairs= pappus or into scales or absent.
Corolla:- Petals (5) tubular
Androecium: Stamens 5, epipetalous (attached to petals by their filament) syngynesions (anthers united but filaments free).
Gynoecium: Carpels (2), ovary interior, 1 – celled with one basal, anatropous ovule, style 1, stigma bifid,
Fruit:- crypsela
Ray floret: Zygomorphic, unisexual or sometimes neuter (e.g sunflower) and epigenous.
Calyx: Modified into pappus, scale or absent
Carolla: Petals (5), gamopetalous, ligulate (strap – shaped)
Gynoecium: as in disc florets

Family Bambacaceae

Characteristics:

This family has the following characteristics:

- **Habit:** These are large trees
- **Leaves:** These are simple or digitately compound with deciduous stipules
- **Flowers:** The flowers are regular, large, bisexual and hypogynous.
- **Calyx:** This is gamosepalous with (5) sepals. It is valvate, often with an epicalyx.
- **Corolla:** This is polypetalous with 5 petals and imbricate
- **Androecium:** These are 5-stamens, free or polyadelphos.The anthers are two celled sometimes more. The pollen grains are smooth and staminodes are often present
- **Gynoecium**: This is syncarpous, with (2-5) carpels. When 5, the carpels are opposite to the petals.
- The ovary is superior, multilocular with 2 - ovules in each loculus.

- **Fruit:** The fruit is capsule
- **Seeds:** These are small, often very heavy and with scanty or no endosperm.
- Eg *Bambax*, red or silk cotton tree, *Bambax ceiba*

Family Rutaceae (citrus)

- **Habit:** These are shrubs and trees
- **Leaves:** The leaves are simple or compound, alternate or rarely opposite and gland dotted
- **Flowers:** These are regular hypogenous and bisexual. The disc below the ovary is prominent and ring or cap like.
- **Calyx:** There are four or five sepals free or connate below and imbricate.
- **Corolla:** petals four or five free imbricate
- **Androecium:** The number of stamens varies. They can be as many or more often twice as many as the petals or numerous as in citrus. They are free or united in irregular bundles and inserted on the disc.
- **Gynoecium:** These are generally (4) or (5) carpels, as in citrus. They are syncarpous or free at the base or united and either or seated on a disc. The ovary is generally four or five locular or multilocular as in citrus, with axile placentation. There are usually two ovules in each loculus arranged in two rows.
- **Fruit:** This is a berry, capsule or hesperidium.
- **Seed:** The seeds may or may not have an endosperm. Polyembryony is frequent in citrus Eg lemon , citron, lime, orange and pummelo.

family meliaceae

- **Habit:** These are mostly trees rarely shrubs.
- **Leaves:** The leaves are pinnately compound and the leaflets are oblique.
- **Inflorescence:** This has an axillary panicle
- **Flowers:** These are regular, often bisexual sometimes polygamous and hypogynous.
- **Calyx:** There are (4-5)sepals (gamosepalous)
- **Corolla:** There are 4-5 petals(usually polypetalous)imbricate
- **Androecium:** There are 8-10 stamens, generally united in to a long or short staminal tube.
- **Gynoecium:** There are (2-5) carpels(syncarpous)the ovary is superior, it is 2-to5-locular,rarely unilocular, with one or two ovules in each chamber. disc is annular surrounding the ovary.
- **Fruit:** The fruit is a capsule, berry or drupe.
- **Seed:** This is often winged, and albuminous eg Mahogany, satin wood.

Family anacardiaceae

- **Habit:** These are shrubs or trees. Many of them are resinous.
- **Leaves:** These are simple or pinnately compound alternate and exstipulate.
- **Inflorescence:** This is a panicle of many small flowers .
- **Flowers:** The flowers are small, regular and bisexual they are sometimes polygamous, hypogynous to epigynous and usually pentamerous. The disc is present.
- **Calyx:** There are usually 5 sepals, though they vary from 3-7.they may be free or united.
- **Corolla:** There are as many petals as sepals. The corolla is sometimes absent. it may be free or connate.
- **Androecium:** There are 10-15 stamens (the number of fertile stamens varies in many cases. They are free and inserted on an annular disc
- **Gynoecium:** These are usually (3-1) carpels and rarely 5 (syncarpous).The ovary is superior or some times inferior .it is often 1-celled (rarely 2-5 celled) and has one ovule in each chamber .often only one ovule matures in to a seed.

• **Seed:** The seed is exalbumenous with a large curved embryo. Eg mango, cashew tree.

Family Sterculiaceae

• **Habit:** These are shrubs or trees rarely herbs.

• **Leaves:** leaves and stipules are like those of Malvaceae.

• **Flowers:** The flowers are often regular and zygomorphic. They are bisexual, rarely unisexual and hypogynous.

• **Calyx and corolla:** These are like those of malvaceae, sometimes corolla is absent. There is no epicalyx.

• **Androecium:** The number of stamen varies from 5-25 though they are usually ∞ stamens. They are mostly arranged in two whorls the outer whorls opposite to d sepal and often reduced to staminodes or absent and the inner whorl opposite to the petals, being fertile and often branched all stamens is more or less united below into a tube or sometimes on the gonophore. The anthers are 2 locular.

• **Gynoecium:** There are (5-2) carpel (often 5),syncarpous. The ovary is superior and in 5- to 2-locular, with 2- anatropous ovules in each loculus. The style is simple and the stigma lobed.

• Fruit: The fruit can be dry or fleshy and is often a schizocarp.

• Seed: This has a fleshy endosperm, sometimes arillate, example cola, cocoa tree.

Technique for collecting and preserving common flora (HERBARIUM)

Herbarium is a collection of dried, pressed, nicely preserved plant specimens arranged in sequence of an accepted system of classification. It is used for identification of fresh specimens, comparison between floras of different regions. The value of a herbarium depends on the care with which specimens were preserved and mounted, accuracy and completeness of identification and labeling and maintenance so that they do not deteriorate and can easily be referred to. Equipment for collection and collecting methods.

1. Specimens that have both flowers and fruits are selected.
2. Digger / trowel – for digging out the underground parts of the plants example Rhizome, corm, tubers etc. These parts help in identification and classification of the plant.
3. Pruning knife/bent sickle:- These have to be sharp. Pruning knife is used for cutting branches of herbs, shrubs and trees. Bent sickle has a long handle and is used for collecting flowers/fruits from high trees.
4. Vasculum/plastic bags:- Vasculum is an oval metallic container with screwed fasteners to keep it airtight to avoid wilting. Plastic bags can also be sued as they prevent loss of moisture when close tightly.
5. Field note Book:- This is a pocket sized note book used to record all information concerning the plant: These include

a. Location e.g. country, province, Stale Park, river. Etc
b. Habitat example river bank, forest edge, open field, swamp etc
c. Field observation of the plant example height flower, colour, fruit colour, odour.
d. Nature of stem juice example milky colour sticky, red, clear sap, thick, watery
e. Collection number example 1 . . . . . followed by your last name
f. Date of collection
g. Name of plant if know name could be local, common or botanical
h. Association with other plants and its abundance.

Plant Press

This consists of equal sized top and bottom frames measuring 30cm x 45cm made of these, light, strong pieces of plywood or metal. Between the frames are the folders, blotters and ventilators. Folders are old newspapers in which the collector places the specimens. Blotters absorb moisture from the specimens while ventilators provide air for drying. The press is tightly bound together by strong straps of canvas or leather and is placed in the hot sun or suspended over moderate heat.

Plant Pressing

Plants for pressing should be placed in the folders in a manner so as to give maximum possible natural look. Longer specimens are bent into a V or W shape so as to fit them on a standard sized mounting sheet (30 x 45cm). Leaves, flowers parts should not be crouded. Thick organs may be trimmed and cotton paid placed proper pressing.

Mounting of pressed specimens:

The pressed, dried specimen are glued or gummed to a hard standard sheet of paper. The paper should be fairly heavy to reduce. Flexing when mounted specimen are picked up.

Label

Label is an important part of finished but usually 6 x 10cm. they are pasted with glue on the lower right hand corner of the mounting sheet. A label should contain the following information's.

1. Region of collection accompanied by name of the institution.
2. Botanical name of the plant with the author citation
3. Locality of the collection
4. The habitat
5. Name of the collector and his field number
6. Collector field observation
7. Date of collection and the name of the person who identified the specimen.

Filling

This is the arrangement of specimens in a herbarium. Specimen are filed according to a particular system of plant classification filling is done alphabetically for genera or according to phylogenetic sequence (in major herbaria). Specimens belonging to are genus are placed within a genus cover. In many major herbaria, the scheme of technicolour folder is in usage. The colour of the folder is indicative of the distribution of the plant.

Care of specimens

Herbaria in warm and moist regions appear to face pest problems. The common pest adults and larvae of tobacco beetle (*Lasioderma serricorne*) and drug store beetle (*Stegobium paniceus*).

These are controlled in the following ways:

1. Use of insecticides example Dichlorodiphenyl) trichloroethane (DDT), carbon disulphide and cyanide gas.
2. Specimens are poisoned with 10% aqueous solution of mercuric chloride
3. Use of insect repellants example Naphthalene balls and Para dichlorobenzene mixed in the ratio 2:1 and put in herbarium case.
4. Use of calcium chloride in small bottles the mouth covered with lose cotton and kept in herbarium cases to absorb moisture thereby preventing fungal attacks.
5. Specimen should be handled with care to avoid cracking and crumbing, tear and wear
6. Store specimens in steel cabinets and fire – proof buildings

Uses of Herbarium

1. The vegetation in any particular region is easily known.
2. While exploring a new area, the herbaria of previously explored areas can be referred to
3. Data about the nature, distribution and ecology of any plant of economic or medicinal value can be obtained easily.
4. It can serve as the basis for other branches of botany example plant ecology. Phytogeography, Economic botany e.t.c

STB III LECTURE NOTES.

FUNGI PLANT AND ANIMAL TAXONOMY

Characteristics and Classification of Plant Kingdom

Algae are reffered to as green thallophytes that have chlorophyll. In other algae some Colours may covered the green Colour, but chlorophyll is always present in all of them.

Algae are authotrophs which means they can manufacture their own food with the help of chlorophyll (chlorophyll is the green substance in plants which help them to absorb light from the sun to enable them grow).

The body of an algae is composed of a true parenchymatous tissue, the cell wall of an algae is composed of true cellulose. Algae live in water or in wet substrata, it also contain a reserve carbonhydrate which usually consist of starch.

Structurally algae may be unicellular, multicellular, filamentous or thalloid. Reproduction in Algae takes place vegetatively by cell division or detachment of a portion of the mother plant, or assexually by spores, or sexually by gametes.

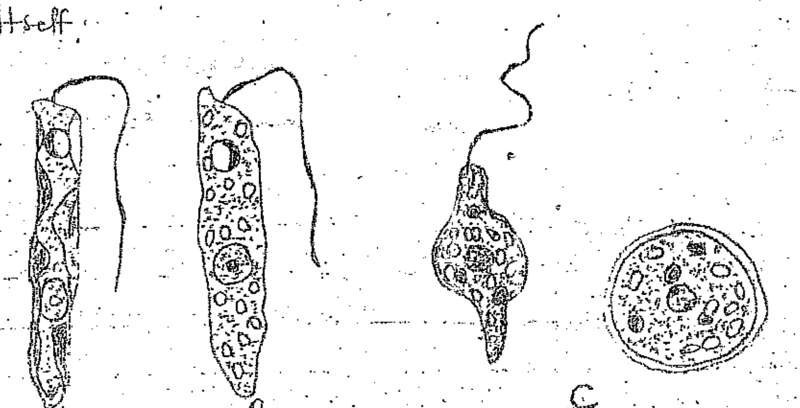
Algae can be grouped into six different classes which are:-

I- Cynophyceae or Myxophyceae which are the blue-green algae they belongs to the small group primitive algae. They are characterized with presence of a blue pigment called Phycocyanin together with the chlorophyll making the blue-green colour, the body is clearly differntiated with protoplast having a simple method of reproduction. Majority of them are fresh water dwellers, and are often found largely in every stagnant pool of water, wet ground or in the form of road slime after rains.

The cell structure is primitive with no definite nucleus with differentiated protoplast. Cynophyceae or myxophyceae or blue green algae have about (1,500 sp) e.g Gleocapsa, Oscillatoria, Nostoc, Anabaena, Rivularia etc.

II- Euglenophyceae it have almost 18 sp. Euglena belongs to the family euglenaceae is one of the most simple unicellular organism. The evolution of higher form of plants started from it, it belongs to the flagellatae group. It belongs to the border of plants and animals kingdom. It is mostly found largely in polluted waters containing organic substance having a bright green colours. It is a single cell, naked, free swimming organism, elongated with one end blunt and the other tapering. It has a single cilium which is long, slender, whip-like projection which helps the plant to swim in waters.

Inside the protoplast their is central nucleus, green plastids, a contractile vacoule which usually contracts and expands in few seconds with a red spots near the blunt-end called the eye spot. Feeding is by photosynthesis. There is no sexual reproduction in euglena, it multiplies by longtitudinal division of by itself.



Different species of euglena.

Fig A - green form of algae B colourless (saprophytic) form. C Various forms (three shown) assumed from a single cell.

III- Chlorophyceae or green algae; they are characterized by presence of chlorophyll located in plastids (chloroplasts). They are mostly fresh water algae with few of the species being terrestrial. They are unicellular or colonial being motile or non-motile. While the multicellular being thalloids or filamentous. Chlorophyceae have a well organised protoplast with definite nucleus. Usually one in each cell or several having one or different chloroplasts.

The chloroplast vary in shape and size depending on the species or genera. They can be cup-shaped, plate-like, stellate, spiral, spherical, oval or discodial bearing one or more pyrenoids. Most unicellular or colonial forms are provided with whip-like structures called cilia often 2, sometimes 4 or many for motility of cells or colonies.

Chlorophyceae have two or more contractile vacoules and a small eye spot

Reproduction. Vegetative reproduction takes place by spores of various types: (a) a motile, ciliate spore (zoospore), (b) non-motile, non-ciliate spore with distinct wall of its own but, produced within a mother cell (abortive zoospore) (c), a vegetative cell acting as a spore having no wall of its own, the wall of mother spore acting as ~~the~~ the wall of spore (modified vegetative cell).

Sexual reproduction takes place by Isogamy, anisogamy or oogamy depending on the species. However the mode of the specie sexual reproduction occur by homothalic (i.e pairing gametes come from the same parent), while others are heterothalic (i.e the pairing gametes come from two seperate parents). In many green algae, it has been observed that a gamete grows parthenogenetically (without fusion with another gamete) into a new plant.

IV Bacillariophyceae or diatoms.

It is commonly called diatoms, it is mostly or one celled algae. These are of infinite variety of forms and often of single cells may occasionally forms filaments and colonies. They are universally distributed in fresh water, as ~~well as the~~ salt water, and also in wet ground.

Most diatoms are free floating, but some are attached by a gelatinous stalk. There are over 5,300 living species of diatoms. Diatoms may be boat-shaped, rod-shaped, disc-shaped, wedge-shaped, spindle-shaped, circular, oval, rectangular, etc.

The wall of diatom cell is made of two halves or valves, one (older) fitting closely over the other younger, very much like a pill-box or soap case. The outer valve is known as epitheca and the inner as the hypotheca. The jerky movements seen only in diatoms with raphes, may be due to the streaming movement of the cytoplasm along the raphe.

The colour is due to the presence of a golden-brown pigment called diatomin, in addition to chlorophyll. Pyrenoids may or may not be present. When present, they do not have a starchy envelope. The reserve food consist of globules of fat and globules of an insoluble complex substances called volutin. No starch is formed in diatoms.

Reproduction. Diatoms may reproduce vegetatively (by cell division usually at night), assexually (by auxospores), and sexually (by conjugation of gametes).

In Vegetative reproduction, the Protoplast grows, resulting in the separation of the two Valves. Each of the two half-cells forms a new value against the old one, fitting into it. Division and value formation proceeds Continously.

Uses

Diatomaceous earth has a variety of uses, It is used in metal polish toothpaste, Paints and Plastics. It is extensively used as a filter in filtration of liquids e.g Sugar refining. It is also used as a heat insolulator in boilers and furnaces.

CLASS PHAEOPHYCEA

Phycophyceae or brown algae are among the Intresting group of seeweeds with a variety of Peculiar forms and sizes Comprising about 1,000 species. They are widely distributed between tidal Level along Sea Coasts, Predominantly of temperate seas. They are attached to the rocks or orther substrata.

In colder regions they go beyond a depth of 20m; while in warmer areas of Seas, a few species may grow to a maximum depth of 90m. A few are free floating. Their colour ranges from brown to olive-green due to the presence in the chloroplast of a brown pigment, fucoxantin, which mask the chlorophyll. There are no Pyrenoids. The reserve food may be Kind of a sugar (and not starch) or more commonly, a Complex Carbonhydrate Called laminarin.

Some like Ectocarpus, are short filaments, while orthers, like fucus and Sagaseum They grow at or below the low tide Level, extending far into the Sea, to a depth of 90m. Phycophyceae (except fucus and Sagassum) show a regular alternation of Generation, with different degrees of development of the Saporophyte and the gametophyte.

Reproduction: Motile reproductive bodies (Zoospores and gametes or Sperms) are Universally Present throughout Phaeophyceae. Several species reproduce Vegetatively by fragmentation of the thallus. Most of them reproduce asexually through zoospores or aplanospores (except Fucus and Sargassum), and Sexually by Isogamy, anisogamy or Oogamy. The Zygote germinates without any Period of rest.

Uses: are used as Source of fertilizer in Coastal areas, Kelps are use as Source of Iodine, they Contain Sugar and Several of them are rich in Vitamins. ~~It is also consumed as food source for fishes in some coastal~~ areas of Japan and China. Most brown Algaes are source of

4 Food to Fishes.

Bryophyta (Liverworts and Mosses)

Bryophytes are the simplest land plants. They originally evolved from green algae. The phylum have two classes namely: the hepaticae or liverworts and the Musci, or mosses.

None of the group is particularly well adapted for life on land they are mostly confined to damp, shady places. Bryophytes are small simple plants with strengthening and conducting tissues are absent and poorly developed.

There is no true xylem or phloem. They don't have true roots, being supported by ~~two~~ thin filaments called rhizoids which grows from the stem. The plant absorbs water and mineral salt by the whole surface including the rhizoids, unlike true roots. The true roots possess vascular tissue, as do true stems and leaves.

Most bryophytes have adapted to survive periods of dryness using mechanisms that are not fully understood.

It has been shown that the well known xerophytic moss Grimmia pulvinata can survive total dryness for longer than a year at 20°C. There will be rapid recovery by the plant as soon water becomes available.

Alternation of generation.

Very common with all plants and some advanced algae, such as Laminaria, bryophytes show alternation of generation in two forms of organism, a haploid gametophyte generation and a diploid saprophyte generation, alternate in the life cycle.

The haploid generation is called gametophyte becouse it undergoes sexual reproduction to produce gametes. Production of gametes involves mitosis, so the gametes are also haploid. The gametes fuse to form diploid zygote which grows into the next generation, the diploid saprophyte generation. It is called the saprophyte becouse it undergoes asexual reproduction to produce spores.

Characteristics of Bryophytes

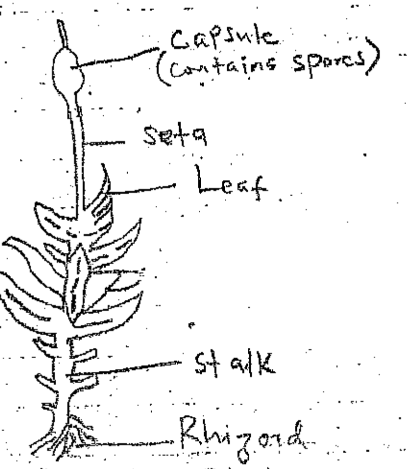
They are advanced multicellular green plants.

Cells are differentiated into tissues.

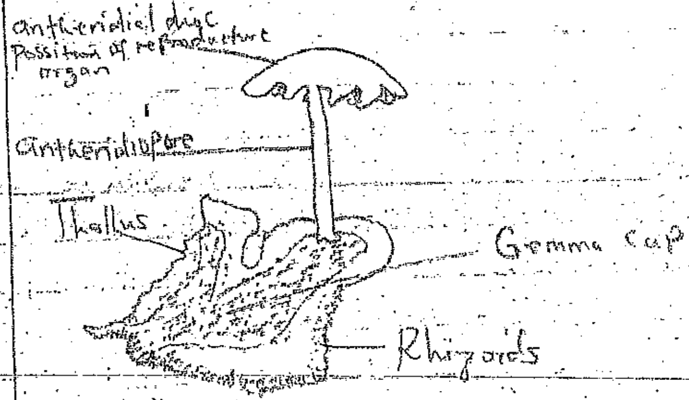
They lack true roots, stems and leaves but have structures resembling roots, stems and leaves.

They are non-vascular plants

v) Commonly found growing in moist places.
(vi) Some bryophytes are terrestrial while others are aquatic
vi) They reproduce assexually by spores with alternation of generation.
vii) Examples include Mosses and Liverworts.



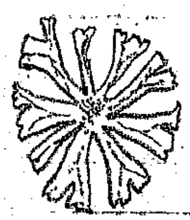
A Moss Plant



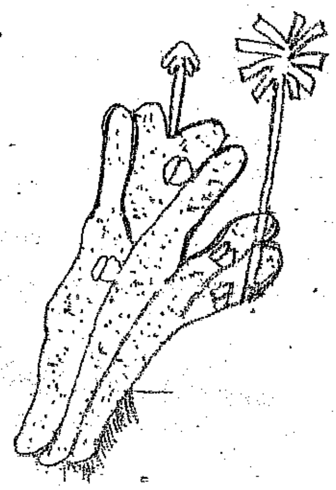
A Liverwort

Bryophyta from the ecological point of view, they play a significant role in soil conservation. Bryophyta comprises land-inhabiting, autotrophic plants which prefer moist and shady habitats.
Bryophytes are widespread in distribution. Members belong to class Hepat-ticae (Liverworts), such as Riccia, Marchantia, Pellia, e.t.c, and class Antherocerotae (horned liverworts) e.g Anthoceros have a dorsiventrally flattened thallus which grows densely, forming a covering like a green carpets on ground. Other leafy members include different kind of moss belonging to class musci, are also gregarious in habit. Wherever they grow, they multiply rapidly due to their

extensive spore-producing capacity and form a continous patch, or a soft velvet-like cover on the substratum.



A Riccia plant.

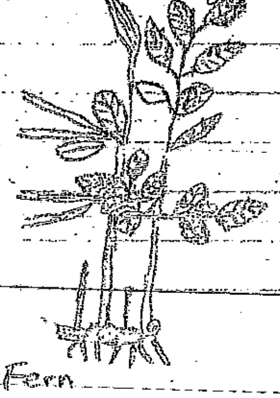


A Marchantia plant.

PTERIDOPHYTA

The origin of Pteridophyta is not certain, it is however, known among the vascular plants the oldest and the most primitive group is Psilophytales which grew aboundantly in the early times. The group soon declined, and before it disapeared (it leaves two genera as its living representatives) giving rise to three evolutionary ~~lines~~ independent lines as represented by Lycopodinae, Equisetinae and Filicinae. Each of them follow its own course of evolution. The origin of Psilophytales is again speculative, it may have been derived from algal ancestors or from some bryophytes.

Pteridophytes are terrestrial large green plants with true roots, stems and leaves. They are non-flowering plants; leaves arise from the rhizomes E.g Ferns which are perenials with underground stems. Reproduction: is true spore formation which is usually dispersed by wind. Sporangium is borne on the leaf in a group called spores example Fern, Dryoteris species, Nephritis e.t.c.



A Fern

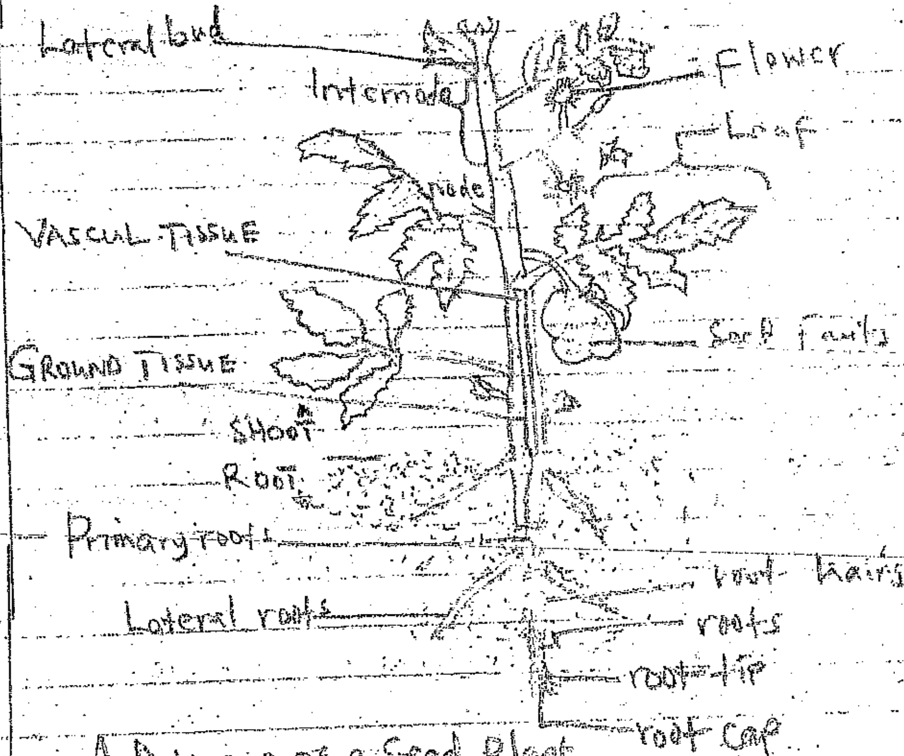
A Dryopteris.

Spermatophytes

The Spermatophytes means" ~~sp~~ seed Plants" are some of the most Important organism on earth. Life on land is shaped largely by the activities of seed plants. Soils, forests, and food are three most apparent products.

Seed-Producing plants are probably the most familiar plants to most People, Unlike mosses, liverworts, horsetails, and most other Seedless Plants. ~~which~~

Many Seed Plants are large. Conifers are Seed plants; they includ Pines, redwood and many other large trees. The other major groups of seed plants are the flowering plants, including plants whose flowers are showy, but also many flowers with reduced flowers such as oaks, iroko, mahogany, grasses and palms.



A Diagram of a Seed Plant

Classification of a Spermatophytes

The spermatophytes also known as (Phanerogams) Comprise those plants that produce seeds. They are Sub-set of the embryophytes or land Plants.

The Living Spermatophytes form five groups namely:

1

- Cycads: a Subtropical group of Plants with a large compound leaves and a stout trunk.
- Ginko, a single living species of trees
- Conifers, Cone bearing trees or Shrubs
- Gnetophytes, woody Plants in the genera Gnetum,
- Angiosperms; the flowering Plants, a large group including many familiar Plants in a wide variety of habitats

Gymnosperms are Plants that do not flower and do not bear Seeds in an enclosure such as a fruit. The Seeds are Produced on a Surface of the sporophylls or similar structures until they are dispersed. The sporophylls are usually arranged in a spiral on the female strobili produce the pollens which will fertilize the ovules in the female cones. The ovule contains a nutritious nucellus which is itself enclosed in several layers, ~~of~~ the layers will eventually become the seed coat, after fertilization and further development of the embryo takes place.

Division: Pinophyta

Subdivision: Pinicae including conifers. Conifers are pine trees as evergreen.

Structure and form - For the Sake of discussion we will look at Pines, which are the largest genus of Conifers. Pine needles are their leaf structures. They are usually arranged in clusters or bundles of two to five Leaves (needles), although some species have as few as one or many as eight leaves in a Cluster, the clusters are sometimes referred to as fascicles.

Some Conifers produce resins in response to injury. The fascicles, needle clusters will fall of every two to five years after maturing, They do not fall off at once unless they are affected by disease.

The Secondary Xylem, wood in Conifers varies in hardness. Most gymnosperms woods consist of tracheids and has no Vessel members or fibres as do flowering trees.

The Secondary Xylem, wood in Conifers varied in hardness. Conifer wood is ~~considered to be softwood, while the~~

wood of broad leaf trees is Considered to be hardwood. The Xylem rings in Conifers are often fairly wide as a result of rapid growth.
Pine trees can be found in all types of environments and ones of opposite extremes.

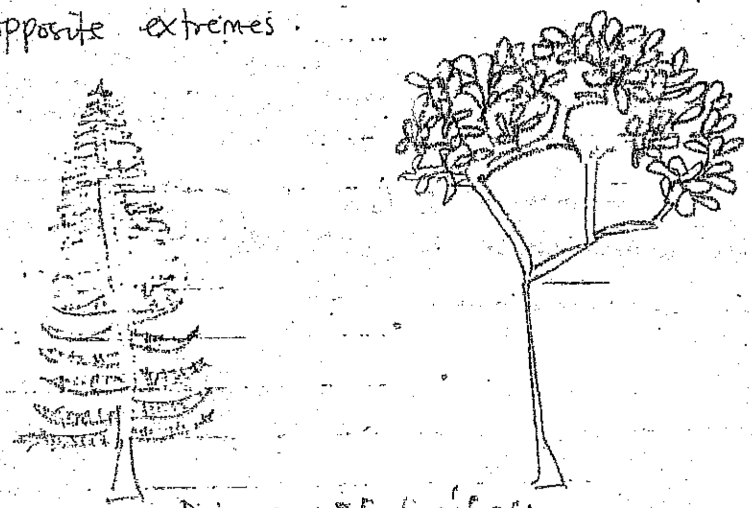


Diagram of Conifers

There are 550 Conifer species because they are well adapted to harsh Climates, they often form the tree line on mountains and in Sub Polar regions.

Class Ginkoatae

Gin Ko trees have Small fan-shaped leaves with veins that evenly fork. They have Similar reproductive cycles to that of Conifers with the exception that the edible Seeds are encased in a fleshy covering. The Covering smells like ranchid butter at seed maturity.

Division Pinophyta

Subdivision Cycadicae

Cycads - these plants look like little Palm trees with un-branched trunks and large Crowns of Pinnately divided Leaves. Their Strobili and Cones are quite Similar to those of Conifers: however their Sperms have numerous flagella Much unlike Conifers.

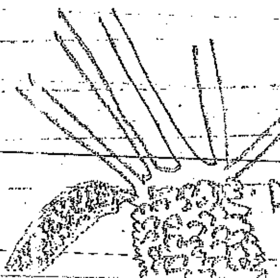


Diagram of Cycads showing cone of Strobili

Differences between gymnosperm and Angiosperm

GYMNOSPERMS	ANGIOSPERMS
1- Xylem exclusively made tracheid	- Xylem mainly compose of resin
2- Phloem Contain no Companion Cells	- Phloem Contain Companion Cells
3 - Flowers are Simple no Calyx, no Corrolla	- Flowers are Complex
4 - Flowers alway Unisexual	- Flowers Unisexual and bisexual.
5- Air Current is the only Pollinating Agent.	5- There are many different Pollinating Agents.
6 - Ovules are freely exposed On the megasporophyll Carpel	- Ovule remain enclosed in the ovary
7 - During Pollination Pollen grain enter the micropyle and are lodged On the nucleus	- Pollen grain are deposited on the Stigma
8 - Male gametophyte presented by a few Cells Usually 2 or 3 Vestigial Prothallus.	- Male gametophyte reduced to 2 nuclei only i.e tube nucleus & generator nucleus
9 - Female gametophyte is a large structure with archegonia embeded in it. Each has Ovum	- Female gametophyte is represented by an 8 nucleate embryo Sac and the Ovum is free in it without any archegonia
10 - Endosperm formed from Vegetative tissue of Female Prothallus before fertilisation and is haploid.	- Endosperm is formed from definitive nucleus only after fertilisation and is (3n) Cotyledons are 1 or 2.